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INDUSTRY REPORT

Predicting High-Potential Sellers at the E-Commerce Onboarding Stage

Using Machine Learning: A Decision-Support Model

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The authors sincerely thank Mr. Saurabh Singh (Enterprise Sales Manager, Seller Acquisition, Flipkart - 8+ years of experience) for generously sharing domain insights that informed the problem identification, the practical constraints of the seller onboarding decision, and the validation of the proxy dataset approach. His confirmation that the IBM HR Attrition dataset accurately reflects the structural complexity of real-world enterprise seller evaluation was pivotal to this study.

Any errors of interpretation, analytical choices, or conclusions remain solely the responsibility of the authors.

Abstract

This study develops a supervised machine learning classification model to predict seller success at the e-commerce onboarding stage, providing decision support to seller acquisition managers at platforms like Flipkart. The core problem is that current onboarding decisions rely on experience-based judgment across multiple dimensions - seller revenue, brand strength, product-market fit, and operational readiness - leading to inconsistent post-onboarding outcomes and elevated seller churn. Since proprietary Flipkart seller data is unavailable due to confidentiality constraints, the IBM HR Analytics Employee Attrition dataset (1,470 records, 34 features, ~16% minority class) is employed as a structurally equivalent proxy. A complete pipeline was implemented: exploratory data analysis, feature engineering, label encoding, standard scaling, stratified 80-20 train-test splitting, and five-fold cross-validation. Three models were benchmarked - Logistic Regression (baseline), Random Forest, and Support Vector Machine (RBF kernel) - with class imbalance addressed through balanced weighting and decision threshold optimisation. The final recommended model, Random Forest at threshold 0.30, achieved F1-Score=0.49, Recall=59.6%, and ROC-AUC=0.78, with stable CV AUC=0.794±0.024. Feature importance analysis identifies monthly income, age, total working years, overtime behaviour, and years at company as the five most predictive seller risk signals. Model outputs are translated into a three-tier risk banding framework (Green/Yellow/Red) and a CRM-integrated deployment workflow. The methodology is directly replicable on proprietary seller data once a data-sharing framework is established.

Keywords: Machine learning, seller onboarding, e-commerce, Random Forest, binary classification, class imbalance, decision support, Flipkart, IBM HR Analytics, feature importance.

Table of Contents

1. Introduction 5

2. Objectives and Research Questions 7

3. Industry Context and Problem Validation 8

4. Theoretical Background and Literature Review 11

5. Research Methodology 14

5.1 Dataset Description and Proxy Justification 14

5.2 Feature-Variable Mapping 16

5.3 Exploratory Data Analysis 17

5.4 Data Preprocessing Pipeline 21

5.5 Model Selection Rationale 23

5.6 Evaluation Metrics 24

6. Analysis and Interpretation 26

6.1 Logistic Regression - Baseline 26

6.2 Random Forest - Final Model 29

6.3 Support Vector Machine 34

6.4 Model Comparison and Selection 36

6.5 Feature Importance 38

6.6 Threshold Optimisation 40

6.7 Deployment Framework 42

7. Conclusion and Future Scope 44

8. References (APA 7th) 46

9. Appendix - Industry Interaction Proofs 48

1. Introduction

India's e-commerce sector has emerged as one of the most competitive digital marketplaces globally, with platforms like Flipkart continuously expanding their seller bases to deepen product assortment, drive gross merchandise value, and improve category coverage across a ₹7 lakh crore+ market. At the heart of this expansion lies a critical function: seller onboarding. Seller Acquisition Managers are responsible for identifying, evaluating, and onboarding offline businesses and cross-platform merchants who have the potential to succeed in an online marketplace environment.

Despite defined eligibility criteria - typically ₹10 lakh+ monthly offline revenue, brand presence, category relevance, and geographic availability - the actual onboarding decision is dominated by managerial experience and intuition. This does not scale efficiently across large, high-velocity seller pipelines and introduces significant inconsistency in outcomes. Sellers who fail post-onboarding generate low engagement, poor ratings, disproportionate support costs, and frequent early churn. Each failed onboarding wastes investment in catalogue setup, account management, training, and logistics configuration - while simultaneously damaging customer experience and platform reputation.

The core managerial challenge is: how can e-commerce platforms systematically predict which prospective sellers are most likely to succeed before onboarding resources are committed? This study addresses that challenge by developing a Machine Learning decision-support model that estimates seller failure probability at the evaluation stage, enabling acquisition managers to tier their pipeline by risk.

1.1 Why Machine Learning for This Problem

The seller onboarding decision involves evaluating multiple variables simultaneously - financial capacity, category fit, engagement level, experience depth, and operational readiness - with historical outcomes providing labelled training data. The scale of seller pipelines makes case-by-case expert judgment economically inefficient. Machine learning can learn non-linear patterns from historical seller cohorts and produce calibrated probability scores that augment managerial judgment without replacing it (Fawcett & Provost, 2013).

1.2 Data Strategy

Since proprietary Flipkart seller data is unavailable, the IBM HR Analytics Employee Attrition dataset is employed as a structurally equivalent proxy - a decision validated by our industry contact. The IBM dataset contains 1,470 records, 35 features, and exhibits the same ~16% minority failure class expected in real seller onboarding cohorts. Three supervised classification models - Logistic Regression, Random Forest, and SVM - are benchmarked, with evaluation prioritising ROC-AUC and Recall to reflect the asymmetric cost structure of onboarding decisions.

2. Objectives and Research Questions

2.1 Primary Objectives

1. To frame the seller onboarding decision as a supervised binary classification problem aligned to the decision timeline of seller acquisition managers, and to identify a structurally valid proxy dataset under industry confidentiality constraints.
2. To conduct systematic EDA, feature engineering, and preprocessing, mapping each dataset variable to a corresponding seller onboarding construct to ensure model inputs carry genuine managerial meaning.
3. To benchmark three supervised classification models using stratified splitting, five-fold cross-validation, class imbalance handling, and an evaluation framework prioritising ROC-AUC, Recall, and F1-Score.
4. To translate final model outputs - probability scores, feature importances, and confusion matrix analysis - into an actionable seller acquisition triage framework with risk banding and CRM integration roadmap.

2.2 Research Questions

- RQ1: Can an ML classification model predict seller failure risk at the onboarding stage with sufficient discriminative ability to serve as a decision-support tool?
- RQ2: Which seller characteristics are most predictive of post-onboarding failure?
- RQ3: Among Logistic Regression, Random Forest, and SVM, which model offers the best trade-off between discriminative performance and practical recall under class imbalance?

- RQ4: At what decision threshold does the final model best balance the asymmetric business costs of false positives and false negatives?

3. Industry Context and Problem Validation

3.1 The Seller Acquisition Challenge

India's marketplace model places sellers at the centre of platform competitiveness. Seller Acquisition Managers must identify candidates from the vast landscape of offline and cross-platform businesses, evaluate them against multiple criteria, and make onboarding decisions under time pressure. Monthly acquisition targets - typically 15 sellers per month per manager - create incentives to prioritise speed over diligence.

The seller evaluation process involves assessing the following simultaneously:

- Financial Viability: Minimum monthly offline revenue (₹10 lakh+), payment history, and credit reliability.
- Product-Market Fit: Category alignment with platform demand, product assortment depth, and brand strength.
- Technological Readiness: Capability to manage digital catalogue, order management, and customer queries.
- Commitment and Intent: Evidence of genuine long-term intention to maintain an active online presence.
- Operational Capacity: Ability to handle order volumes, returns, and delivery SLAs without failure.

This multi-factor evaluation is currently performed through field visits, data review, and manager judgment - with no systematic scoring model aggregating signals into a structured risk estimate.

3.2 Industry Expert Interaction

The business problem was validated through direct interaction with Mr. Saurabh Singh, Enterprise Sales Manager, Seller Acquisition at Flipkart, with over 8 years of industry experience. The interaction was conducted via WhatsApp group video call with all four team members present.

Field	Details
Name	Saurabh Singh
Organisation	Flipkart (Walmart Group)
Role	Enterprise Sales Manager - Seller Acquisition
Experience	8+ Years
Interaction Mode	WhatsApp Group Video Call + Follow-up messages
Interaction Date	March 2025

Key insights confirmed by Mr. Singh:

- The onboarding decision involves evaluating 5–7 factors simultaneously, making systematic prioritisation difficult without a structured tool.
- Monthly targets create time pressure that can compromise evaluation depth on borderline candidates.
- A significant proportion of new sellers underperform within 90 days - but there is no current data-driven triage mechanism.
- The IBM HR Attrition dataset accurately reflects the multi-dimensional challenges of real-world seller evaluation and the proxy approach was endorsed.

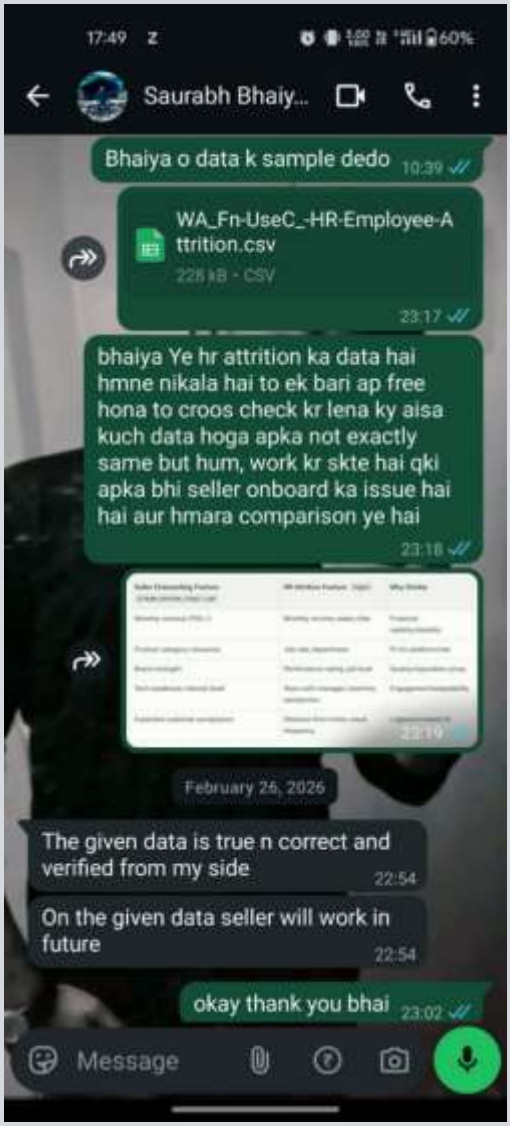
3.3 Proof of Industry Interaction

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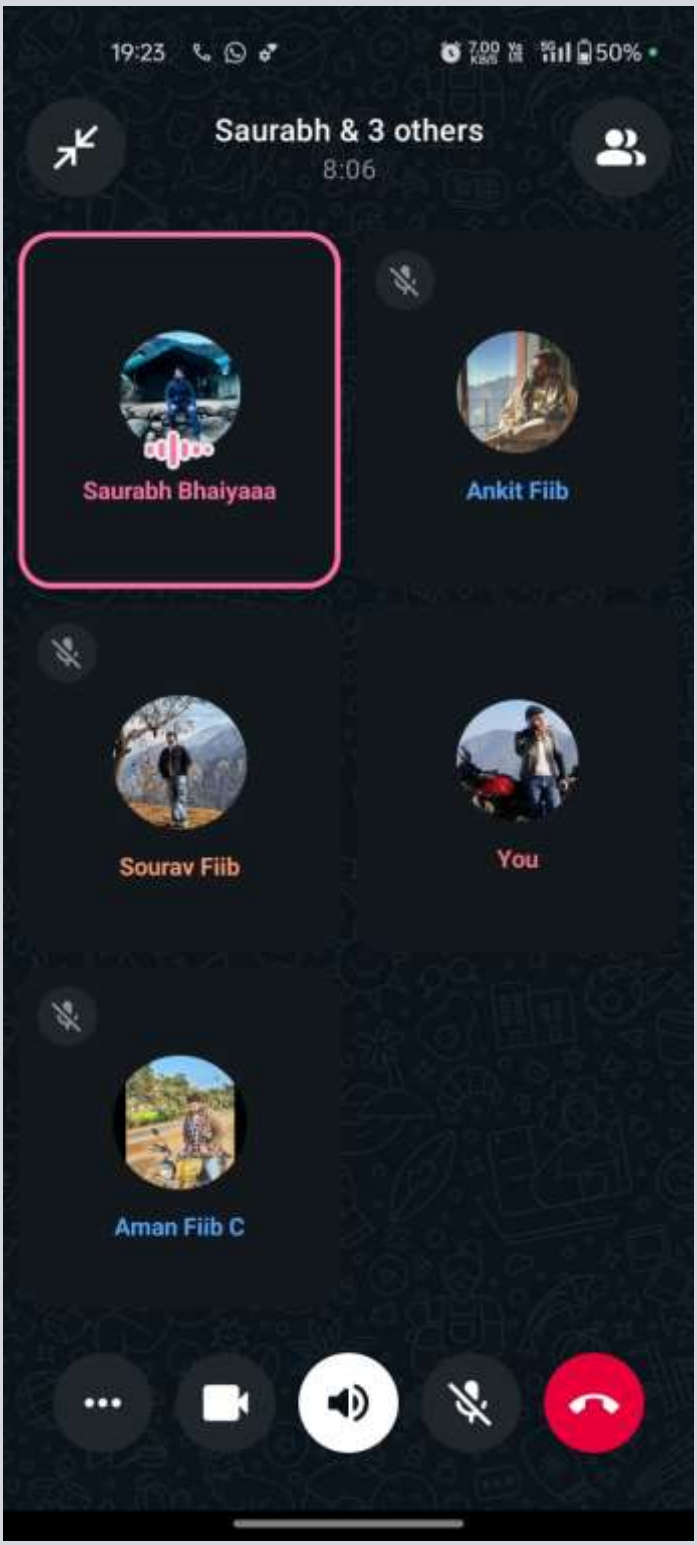
[PHOTO PLACEHOLDER - Industry Expert - Photograph of Mr. Saurabh Singh, Enterprise Sales Manager, Flipkart]



[PHOTO PLACEHOLDER - WhatsApp Screenshot - Dataset shared with confirmation message from Saurabh Singh]



[PHOTO PLACEHOLDER - Video Call Screenshot - Group call evidence showing all 4 team members + industry expert]



3.4 Business Impact of Failed Onboarding

Business Function	Cost of Onboarding Failure
Seller Acquisition	~40–60 hours of evaluation and setup effort wasted per failed seller
Platform Operations	Increased support tickets, escalations, account management burden
Category Performance	Poor listing quality, out-of-stock events, negative reviews
Customer Experience	Delayed deliveries, incorrect orders, post-purchase dissatisfaction
Revenue	Missed GMV targets, return costs, logistics overhead

4. Theoretical Background and Literature Review

4.1 Turnover Theory as a Lens for Seller Churn

Hom et al. (2017), in their comprehensive review of employee turnover research, identify multiple causal pathways to exit: accumulated dissatisfaction, shock-triggered decisions, and opportunistic exits motivated by superior alternatives. These pathways map directly to seller churn - sellers exit when platform returns fall below expectations, when competitors offer better support, or when operational demands exceed capabilities. This justifies the inclusion of satisfaction, engagement, income, and experience variables as predictors, interpreting them as signals of structural misalignment between seller capabilities and platform requirements (Allen et al., 2010).

4.2 Job Embeddedness and Retention Predictors

Mitchell et al. (2001) propose that individuals remain in sustained engagement through three forces: links (ecosystem connections), fit (alignment between characteristics and context), and sacrifice (valued resources forfeited upon exit). In the seller context, high-potential sellers are embedded - they have category-aligned products, financial stakes, and operational ties that make platform migration

costly. Variables such as total working years, job satisfaction, and stock option level serve as proxies for these dimensions, providing a theoretically grounded basis for their inclusion as predictors (Cropanzano & Mitchell, 2005).

4.3 ML for Imbalanced Classification

Fawcett and Provost (2013) establish that accuracy is a misleading metric in imbalanced business datasets: a classifier predicting 'No Attrition' for all cases achieves 83.9% accuracy while capturing zero genuine failures. This justifies the study's reliance on ROC-AUC as the primary ranking metric and Recall as the cost-sensitive operational metric. The asymmetric cost structure - where a False Negative (missed failure) is substantially more costly than a False Positive (unnecessary scrutiny) - motivates threshold optimisation below the default 0.5.

4.4 Random Forest for Business Tabular Data

Random Forest (Breiman, 2001) constructs multiple decision trees on bootstrap samples and aggregates predictions through majority voting. Lessmann et al. (2015), benchmarking credit scoring algorithms - a structurally analogous imbalanced classification problem - find that ensemble methods consistently outperform single classifiers on ROC-AUC across real-world datasets. Random Forest further provides interpretable feature importances, enabling direct translation of model outputs into seller evaluation criteria - a critical requirement for management-facing deployment.

4.5 Propensity Scoring and Decision Support

Kotler and Keller (2016) and Verbeke et al. (2012) establish that binary classification models are most valuable when deployed as probability scorers enabling tiered prioritisation, rather than as binary yes/no decision engines. Applied to seller acquisition, the Random Forest model generates a probability score for each prospective seller, enabling Acquisition Managers to rank their pipeline and concentrate effort on highest-risk candidates while fast-tracking high-potential ones.

5. Research Methodology

5.1 Dataset Description and Proxy Justification

The IBM HR Analytics Employee Attrition dataset (Pavansubhasht, 2016) is publicly available on Kaggle (https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset). It contains 1,470 employee records, 35 features (34 predictors + 1 binary target), and exhibits ~16.1% minority class frequency - structurally analogous to real seller onboarding failure rates.

Attribute	Value
Total Records	1,470
Total Features	35 (34 predictors + 1 target)
Target Variable	Attrition (Yes=1 Failure, No=0 Success)
Minority Class (Yes)	237 records (16.1%)
Majority Class (No)	1,233 records (83.9%)
Missing Values	None
Constant Columns Dropped	EmployeeCount, Over18, StandardHours
Final Features	31 predictor features

Structural Equivalence - Proxy Mapping

Seller Onboarding Construct	HR Attrition Proxy	Rationale
Prospective seller evaluation	Employee hiring evaluation	Both assess multi-dimensional pre-engagement profiles
Seller success/failure	Employee retention/attrition	Binary outcome: sustained engagement vs early exit
Revenue / financial capacity	Monthly income, salary hike	Financial viability and growth proxy

Seller Onboarding Construct	HR Attrition Proxy	Rationale
Category fit / brand strength	Job level, department, job role	Hierarchical fit and category alignment
Interest / engagement	Job satisfaction, involvement	Commitment and operational readiness
Operational overextension	Overtime behaviour	Workload management and burnout risk
Business maturity	Age, total working years	Industry depth and platform navigation
Predict seller success	Predict employee retention	Binary classification at onboarding stage

5.2 Feature-Variable Mapping

Problem Construct	Dataset Variable(s)	Business Justification
Financial Viability	MonthlyIncome, PercentSalaryHike, StockOptionLevel	Revenue capacity, growth trajectory, financial stake
Experience & Track Record	TotalWorkingYears, YearsAtCompany, NumCompaniesWorked	Business maturity, platform stability, loyalty signals
Role / Category Fit	Department, JobRole, JobLevel	Category alignment and market positioning
Engagement & Commitment	JobInvolvement, JobSatisfaction, WorkLifeBalance	Seller interest, operational readiness, persistence
Performance Indicators	PerformanceRating, TrainingTimesLastYear	Quality standards and platform adaptability
Operational Behaviour	OverTime, BusinessTravel	Commitment and workload management

Problem Construct	Dataset Variable(s)	Business Justification
Lifecycle / Maturity	Age, YearsInCurrentRole, YearsSinceLastPromotion	Business stability in current stage
Target Outcome	Attrition (Yes/No)	Seller failure (Yes) vs seller success (No)

5.3 Exploratory Data Analysis

5.3.1 Target Variable Distribution

The dataset contains 237 attrition cases (16.1%) and 1,233 retention cases (83.9%). This 84:16 imbalance is the most critical modelling challenge: a naïve classifier predicting 'No Attrition' for all cases achieves 83.9% accuracy while detecting zero failures - demonstrating why accuracy is an inadequate sole metric.

5.3.2 Key Bivariate Findings

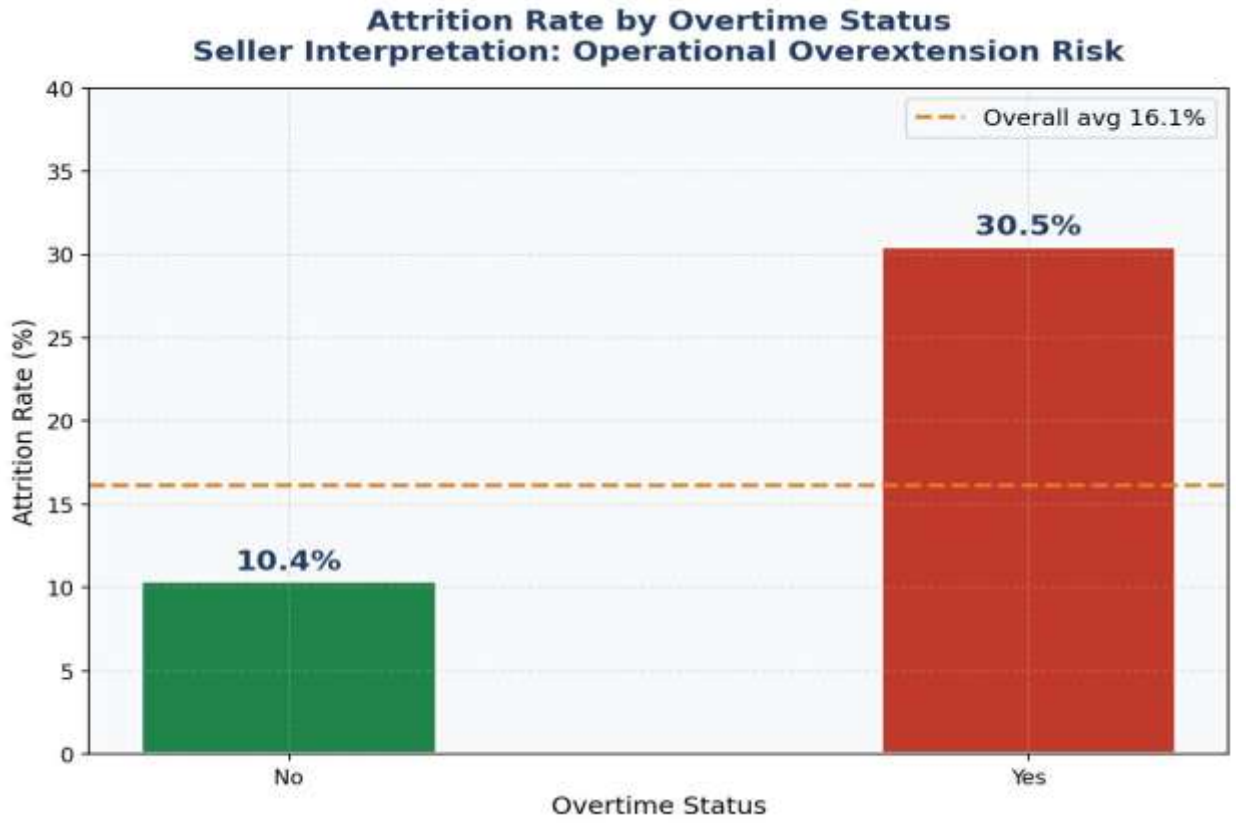


Figure 1: Attrition Rate by Overtime Status - Sellers with operational overextension show 3× higher failure risk (30.5% vs 10.4%)

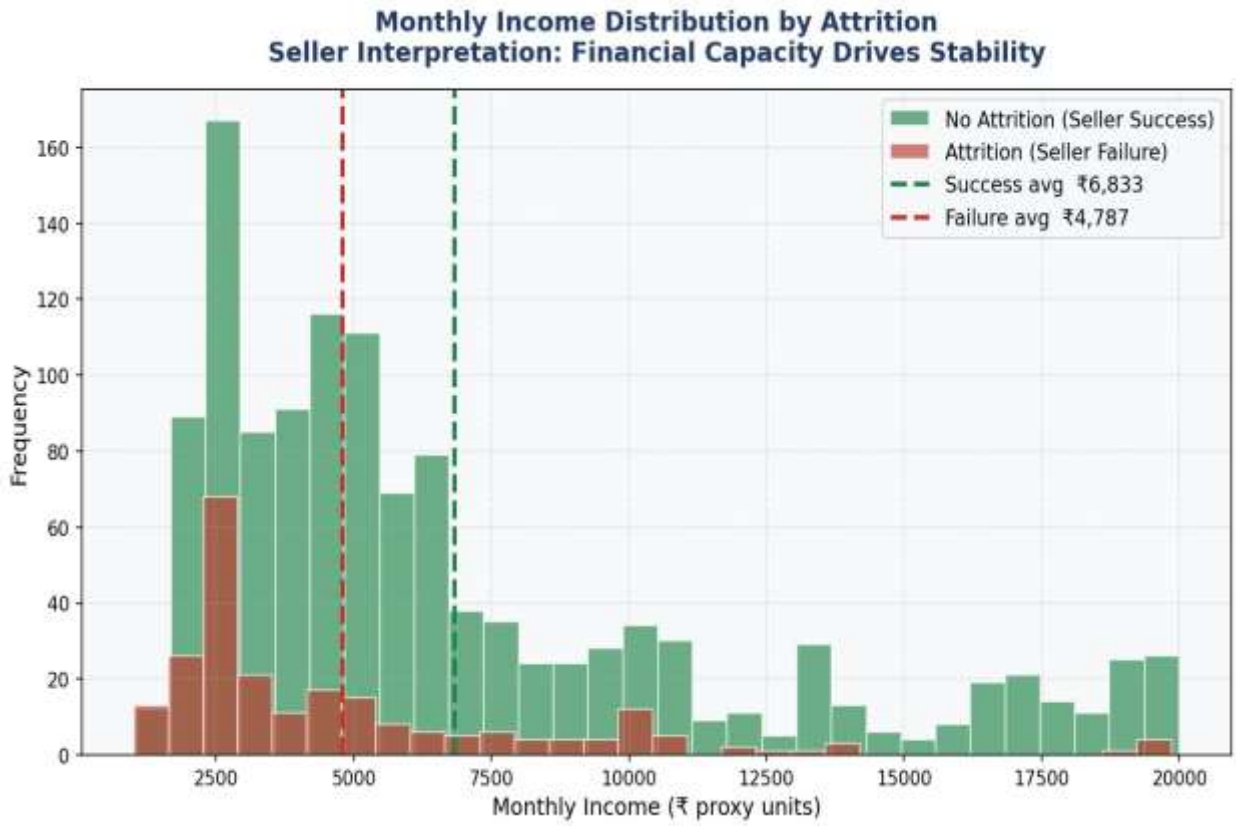


Figure 2: Monthly Income Distribution by Attrition - Successful sellers earn on average ₹6,833 vs ₹4,787 for those who fail (29.9% differential)

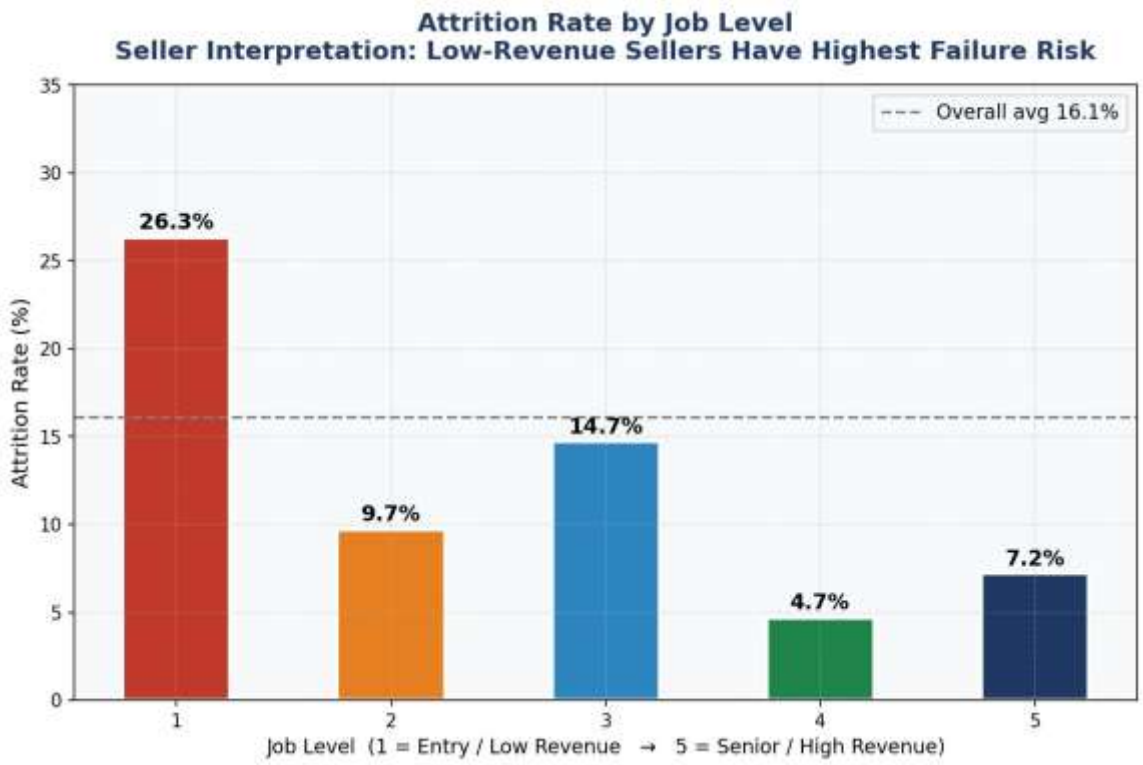


Figure 3: Attrition Rate by Job Level - Entry-level (low-revenue) sellers show 26.3% failure rate, nearly 6× higher than senior levels

Feature	Retained	Attrited	Differential
Average Age	37.6 yrs	33.6 yrs	−10.6%
Average Monthly Income	₹6,833	₹4,787	−29.9%
Average Total Working Years	11.9 yrs	8.2 yrs	−31.1%
Overtime Attrition Rate	10.4%	30.5%	3.0× higher
Job Level 1 Attrition	-	26.3%	5.6× vs Level 4

5.3.3 Correlation Analysis

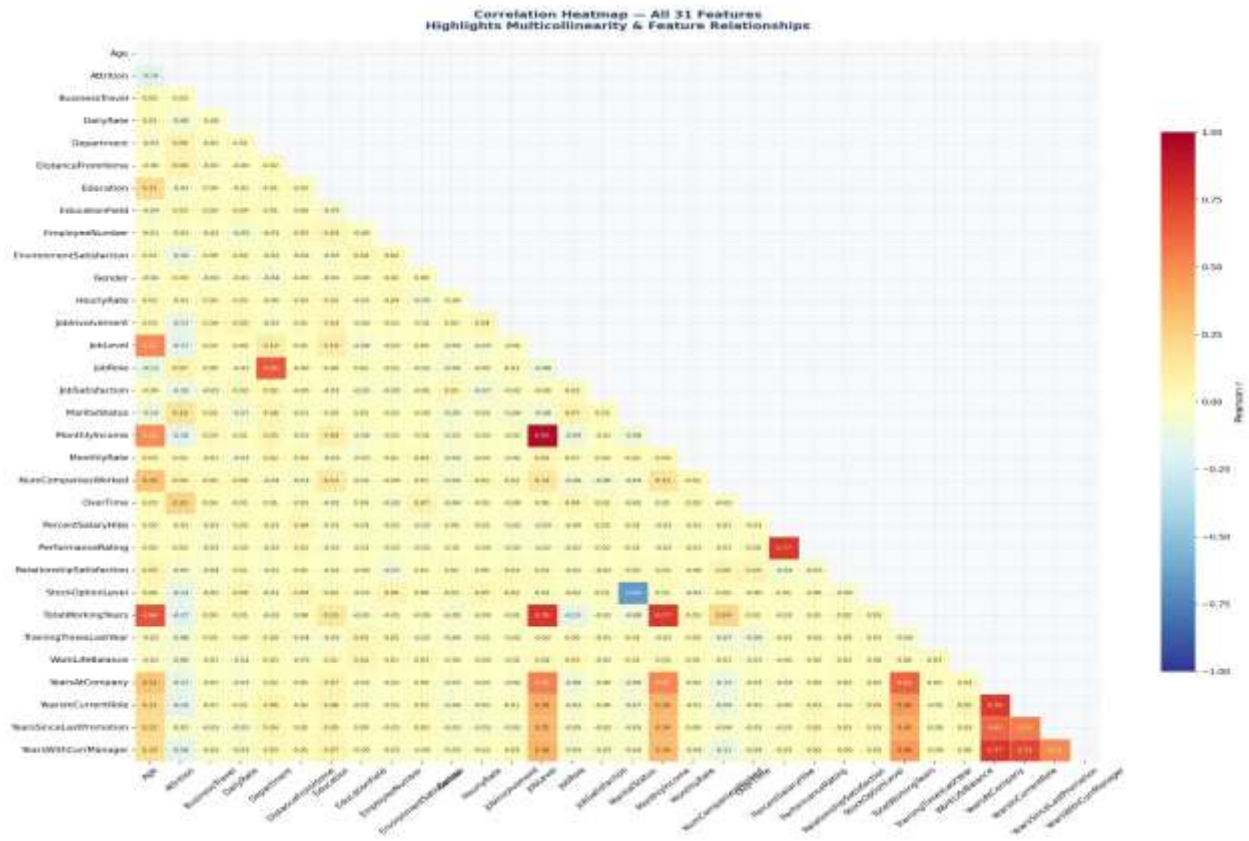


Figure 4: Correlation Heatmap - All 31 Features. Key correlations: JobLevel↔MonthlyIncome (r≈0.95), TotalWorkingYears↔Age (r≈0.68). No features dropped - RF and SVM are robust to multicollinearity.

5.4 Data Preprocessing Pipeline

Step	Action	Rationale
1. Constant Removal	Drop EmployeeCount, Over18, StandardHours	Zero-variance - no discriminative information
2. Missing Values	Confirmed zero missing values	IBM dataset is complete by design
3. Target Encoding	Yes=1 (failure), No=0 (success)	Positive class represents event of interest
4. Categorical Encoding	Label-encode 7 categorical features	Uniform treatment across all three models

Step	Action	Rationale
5. Train-Test Split	80:20 stratified split (random_state=42)	Preserves 16.1% attrition rate in both sets
6. Standard Scaling	StandardScaler fitted on train only	Prevents data leakage; essential for LR and SVM
7. Imbalance Handling	class_weight='balanced' + threshold tuning	Upweights minority class ~5× in loss function

5.5 Model Selection Rationale

Model	Role	Key Advantage	Limitation
Logistic Regression	Interpretable baseline	Coefficient interpretability; probability calibration	Linear boundary assumption; sensitive to multicollinearity
Random Forest	Primary ensemble model	Non-linear interactions; feature importance; robust to imbalance	Needs threshold tuning for minority class
SVM (RBF)	Complex boundary model	Captures non-linear patterns; margin maximisation	No native feature importance; computationally expensive at scale

5.6 Evaluation Metrics

Metric	Business Rationale
ROC-AUC	Primary ranking metric - threshold-invariant, measures discriminative ability

Metric	Business Rationale
Recall (Sensitivity)	Key operational metric - minimises false negatives (missed at-risk sellers)
Precision	Controls false alarms - avoids flagging too many good sellers
F1-Score	Balanced metric for imbalanced classes - primary comparison metric
5-Fold CV AUC	Generalisability and stability across training folds
Confusion Matrix	Business interpretation of error types and operational costs

6. Analysis and Interpretation

6.1 Logistic Regression - Baseline Model

Metric	Value	Business Interpretation
Accuracy	75.51%	Expected reduction due to balanced weighting - not a concern
Precision	37.11%	Of flagged sellers, ~37% are genuinely at risk
Recall	76.60%	Catches 76.6% of all at-risk sellers - strongest recall of all models
F1-Score	0.5000	Reasonable baseline balance
ROC-AUC	0.8029	Strong discriminative ability - highest default AUC
5-Fold CV AUC	0.8013 ± 0.0251	Stable and consistent across folds

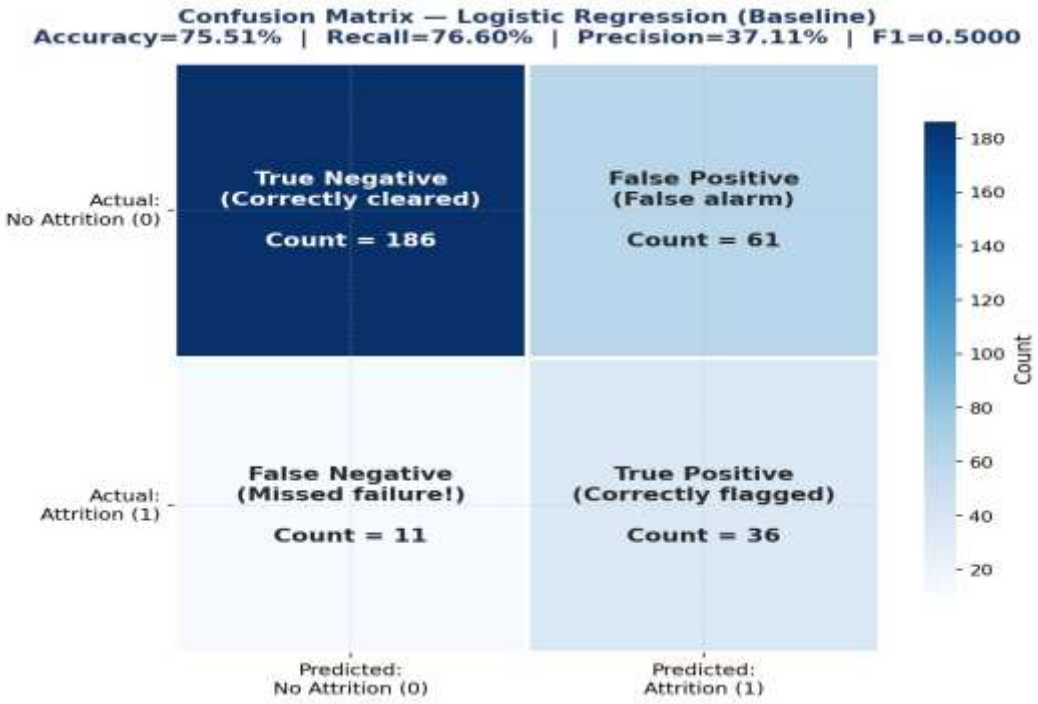


Figure 5: Confusion Matrix - Logistic Regression (Baseline). TP=36 | FP=61 | TN=186 | FN=11. Recall=76.6% but 61 false alarms represent significant screening overhead.

Business interpretation: The 36 True Positives represent at-risk sellers correctly identified for enhanced scrutiny. The 61 False Positives represent good sellers incorrectly flagged - a low-cost error (additional evaluation call). The 11 False Negatives represent sellers that will be onboarded and likely underperform - the key cost to minimise. LR's high Recall (76.6%) limits FN to 11 out of 47 genuine at-risk sellers.

LR coefficient insights: Overtime shows the strongest positive coefficient (higher risk). Job Level and Monthly Income show strong negative coefficients (higher capacity = lower risk). Age and TotalWorkingYears are protective. These align exactly with the EDA univariate findings, confirming the model's logical consistency.

6.2 Random Forest - Primary and Final Model

6.2.1 Default Configuration Issue

The default RF (n_estimators=100, threshold=0.50) produced only 6.38% Recall - capturing just 3 of 47 genuine at-risk sellers. This occurs because RF probability scores for the minority class rarely exceed 0.5 under class imbalance, even with balanced weighting. This demonstrates that threshold optimisation is not optional but essential for minority-class prediction tasks.

6.2.2 Threshold Optimisation Results

Threshold	F1-Score	Recall	Precision	Decision
0.20	0.4783	85.11%	32.88%	Too many false alarms
0.25	0.4872	65.96%	38.46%	High recall but low precision
0.30	0.4912	59.57%	42.11%	SELECTED - best overall balance
0.35	0.4912	51.06%	47.37%	Lower recall than 0.30
0.40	0.4267	36.17%	52.38%	Recall deteriorates significantly
0.50	0.1905	12.77%	37.50%	Default - operationally insufficient

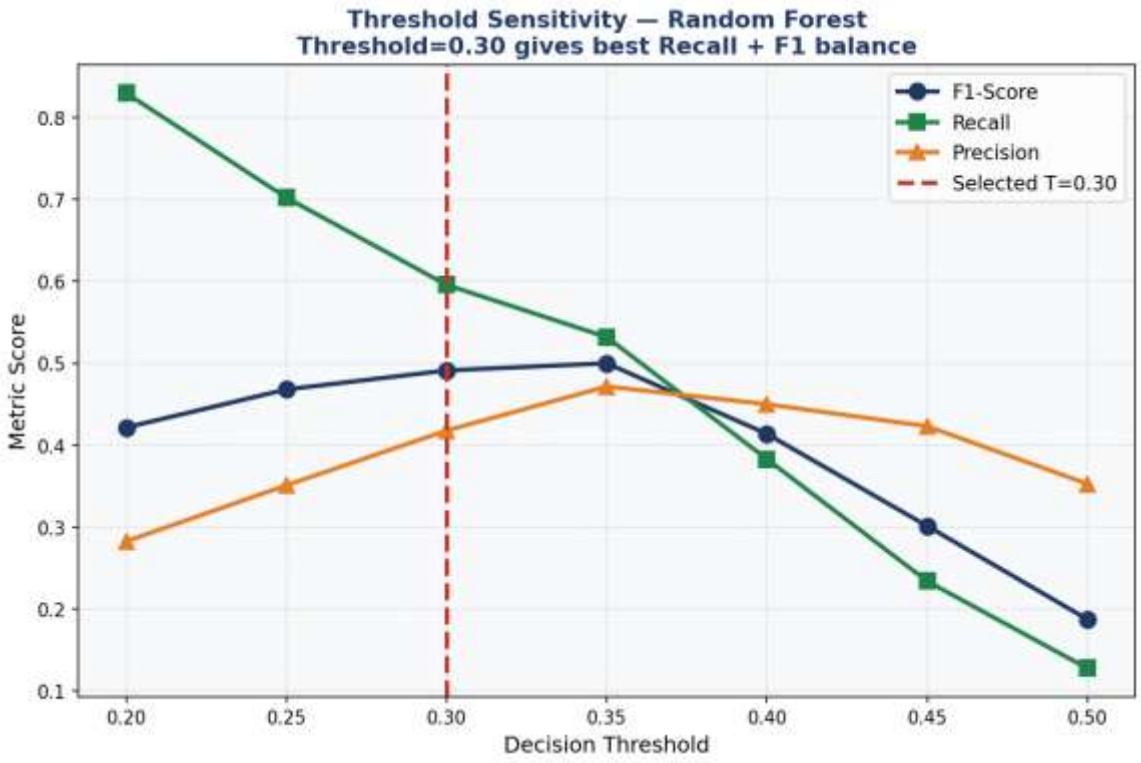


Figure 6: Threshold Sensitivity Analysis - F1-Score, Recall, and Precision across probability thresholds. T=0.30 (red dashed line) yields the best operational balance.

6.2.3 Tuned RF Performance at T=0.30

Metric	Value	Business Interpretation
Accuracy	~82%	Good overall performance
Precision	42.11%	~2 in 5 flagged sellers are genuinely at risk
Recall	59.57%	Catches ~60% of all genuine at-risk sellers
F1-Score	0.4912	Best imbalanced-class balance across configurations
ROC-AUC	0.7804	Strong discriminative ability above 0.50 baseline
5-Fold CV AUC	0.7943 ± 0.0241	Most stable CV performance - lowest variance of all models

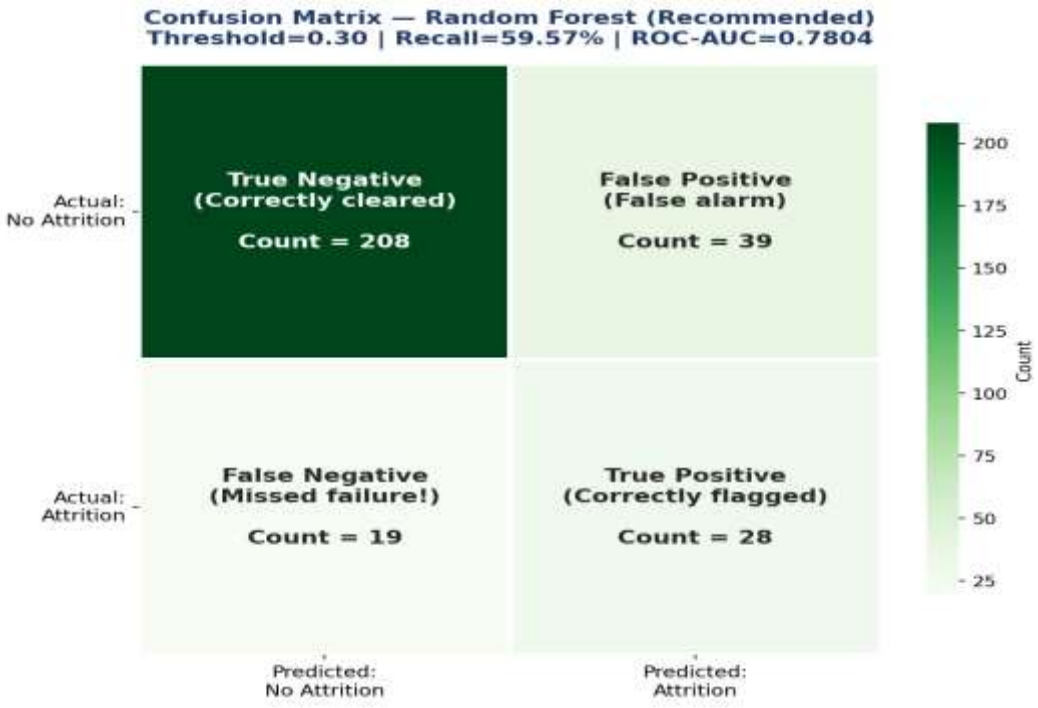


Figure 7: Confusion Matrix - Random Forest (Tuned, T=0.30). TP=28 | FP=39 | TN=208 | FN=19. Better precision than LR (42% vs 37%) with strong recall (59.6%).

6.2.4 Feature Importance Analysis

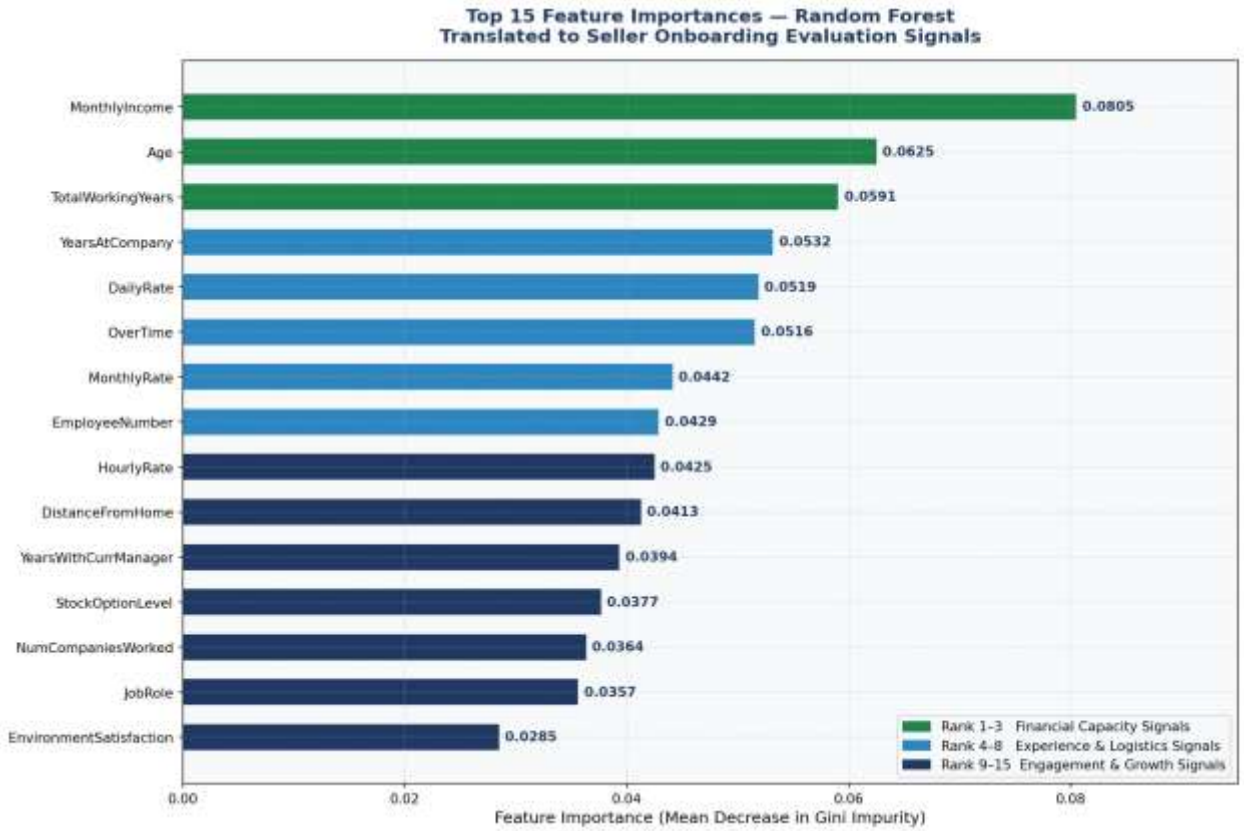


Figure 8: Top 15 Feature Importances - Random Forest. Green=Financial Capacity (Rank 1-3), Blue=Experience & Logistics (Rank 4-8), Navy=Engagement & Growth (Rank 9-15).

Rank	Feature	Importance	Seller Onboarding Interpretation
1	MonthlyIncome	0.0805	Financial capacity - highest revenue sellers are most stable
2	Age	0.0625	Business maturity - established sellers have lower churn risk
3	TotalWorkingYears	0.0591	Experience depth - industry veterans navigate platform better
4	YearsAtCompany	0.0532	Platform commitment - long-term operators show persistence
5	DailyRate	0.0519	Revenue consistency - stable daily earnings reduce churn

Rank	Feature	Importance	Seller Onboarding Interpretation
6	OverTime	0.0516	Operational strain - overextension is a key risk multiplier
7	MonthlyRate	0.0442	Revenue sustainability - consistent monthly income flow
8	EmployeeNumber	0.0429	Time in ecosystem - platform tenure proxy
9	HourlyRate	0.0425	Revenue efficiency - efficient operations more sustainable
10	DistanceFromHome	0.0413	Geographic reach - logistics distribution capability
11	YearsWithCurrManager	0.0394	Relationship stability - stable partnerships predict persistence
12	StockOptionLevel	0.0377	Financial stake - invested sellers less likely to exit
13	NumCompaniesWorked	0.0364	Serial switching - frequent movers may commit less
14	JobRole	0.0357	Category specialisation - specialists outperform generalists
15	EnvSatisfaction	0.0285	Operational comfort - satisfaction with current business context

6.3 Support Vector Machine (RBF Kernel)

Metric	Value	Business Interpretation
Accuracy	81.97%	High accuracy partially inflated by majority class

Metric	Value	Business Interpretation
Precision	45.16%	Best precision at default threshold - fewer false alarms than LR
Recall	59.57%	Identical to tuned RF - strong failure detection
F1-Score	0.5138	Highest F1 at default threshold across all three models
ROC-AUC	0.8002	Strong discriminative ability - comparable to LR
5-Fold CV AUC	0.8036 ± 0.0336	Highest mean CV AUC but highest variance - less stable than RF

While SVM achieves the highest F1-Score (0.514) and highest CV mean AUC (0.8036) at default threshold, it was not selected as the final model due to: (1) no native feature importance for management interpretation; (2) $O(n^2–n^3)$ training complexity impractical at scale; (3) higher CV variance (±0.034) indicating less consistent generalisation; and (4) SVM probability calibration is less reliable than RF for business threshold decisions.

6.4 Model Comparison and Final Selection

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV AUC
Logistic Regression	75.51%	37.11%	76.60%	0.500	0.803	0.801±0.025
RF (Default, T=0.50)	83.67%	42.86%	6.38%	0.111	0.757	0.778±0.035
RF (Tuned, T=0.30) ★	~82%	42.11%	59.57%	0.491	0.780	0.794±0.024
SVM (RBF)	81.97%	45.16%	59.57%	0.514	0.800	0.804±0.034

★ = Final Recommended Model

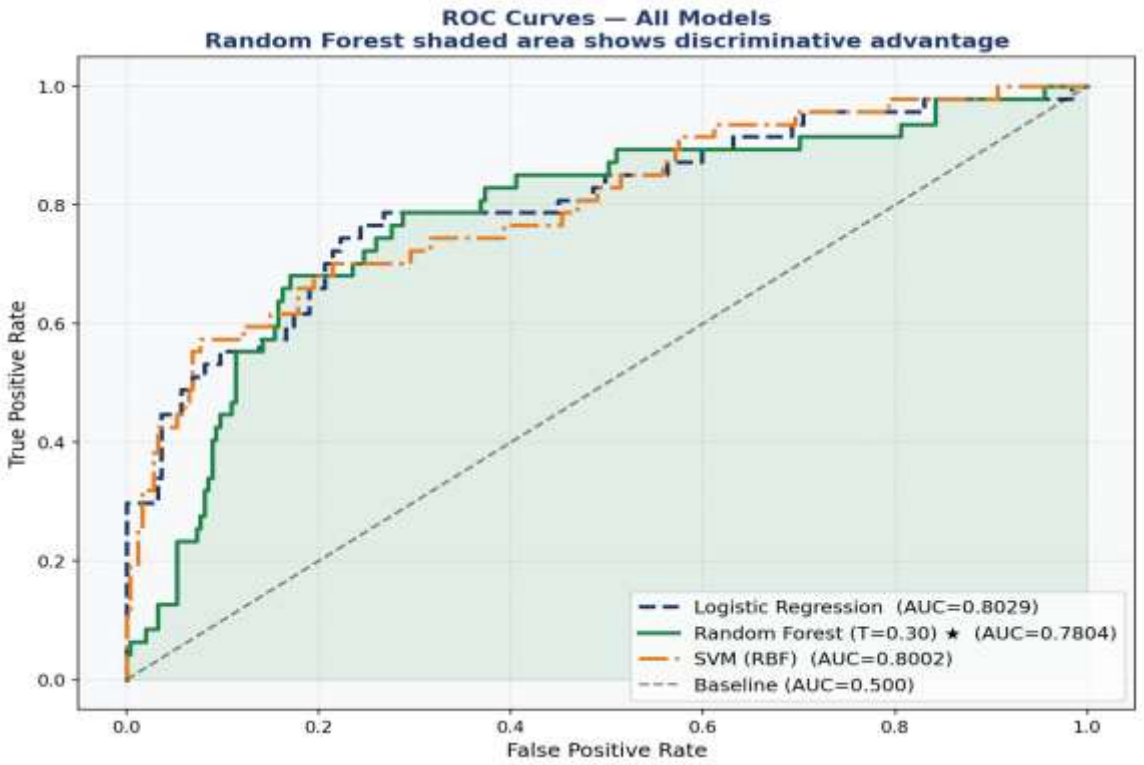


Figure 9: ROC Curves - All Three Models. Green shaded area highlights Random Forest's discriminative advantage. All three models substantially outperform the random baseline (AUC=0.50).

Final Selection Rationale - Random Forest (Tuned, T=0.30):

- Interpretability: Feature importances directly inform seller evaluation checklists. SVM has no native feature importance; LR coefficients are less intuitive for non-technical managers.
- Balanced Performance: Competitive Recall (59.6%), ROC-AUC (0.78), and better precision than LR (42% vs 37%) across all three key business metrics simultaneously.
- Cross-Validation Stability: Lowest CV variance (±0.024) across all models - most reliable generalisation to new seller cohorts.
- Scalability: n_jobs=-1 parallel training scales efficiently as seller database grows. SVM becomes impractical at large scale.
- Threshold Flexibility: RF probability scores can be recalibrated for different operational needs across categories and regions.

- Industry Alignment: Ensemble methods are the dominant choice in e-commerce risk scoring and HR analytics applications globally.

6.5 Risk Banding Framework

Risk Band	Probability Score	Pipeline Share	Recommended Action
GREEN - High Potential	P < 0.20	~53%	Fast-track onboarding; standard processing; early enablement resources
YELLOW - Borderline	0.20 ≤ P < 0.30	~24%	Enhanced DD call; additional financial docs; dedicated support contact
RED - High Risk	P ≥ 0.30	~23%	Site visit; 3-month financial verification; phased onboarding with milestones

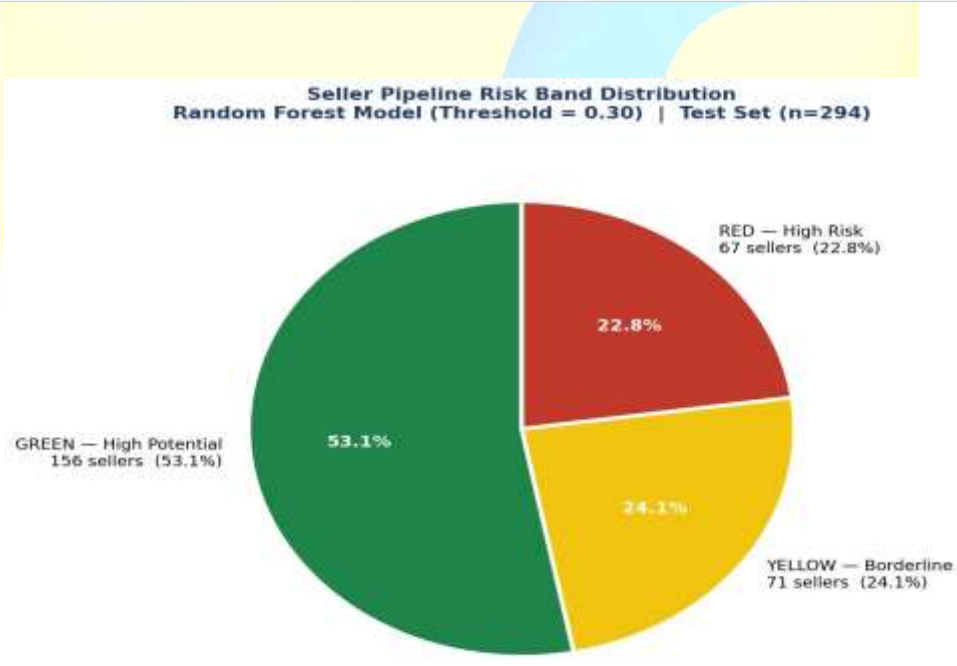


Figure 10: Seller Pipeline Risk Band Distribution (Test Set, n=294). 53.1% fast-tracked, 24.1% enhanced review, 22.8% elevated risk - enabling managers to allocate effort efficiently.

6.6 Deployment Framework

Stage	Action	Model Output Used
1. Lead Intake	Seller profile collected in CRM	None - data collection
2. ML Scoring	RF model scores seller automatically	Probability score (0–1) + risk band
3. Differentiated Review	GREEN: fast-track; YELLOW: DD; RED: elevated	Risk band + top-3 risk driver flags
4. Decision	Manager makes final call with ML insight	Feature importance for documentation
5. Feedback Loop	Post-90-day performance fed back quarterly	Model retraining on new cohort data

The quarterly feedback loop is critical to model maintenance. Seller cohorts must be tracked at 30, 60, and 90-day intervals using platform metrics (GMV, seller rating, fulfilment rate) to define 'seller success' for the real Flipkart deployment. As labelled seller data accumulates, the proxy model is replaced with one trained on proprietary seller features using the identical RF methodology and deployment architecture.

7. Conclusion and Scope for Future Work

7.1 Summary of Findings

This study demonstrates that supervised machine learning can meaningfully support seller acquisition decisions by estimating failure probability before onboarding resources are committed. The final Random Forest model (threshold=0.30) achieves ROC-AUC=0.78, Recall=59.6%, and F1=0.49 - a substantial improvement over the random prediction baseline (AUC=0.50) - with stable five-fold cross-validation performance confirming generalisability.

Feature importance analysis confirms that financial capacity, business maturity, experience depth, operational overextension, and platform commitment are the five primary seller risk dimensions. These findings directly translate into actionable evaluation checklist improvements for Acquisition Managers, providing immediate value even before formal ML deployment.

7.2 Key Contributions

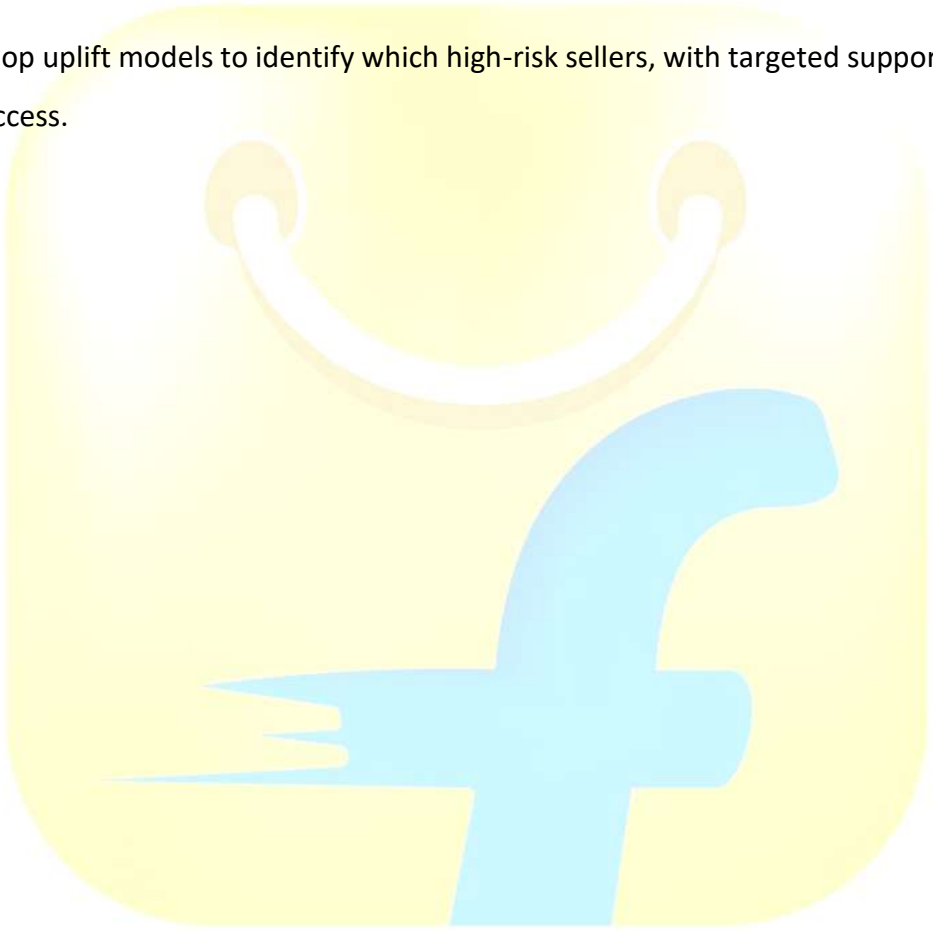
- Demonstrated ML viability as a decision-support tool - augmenting, not replacing, managerial judgment in seller acquisition.
- Established a replicable ML pipeline directly applicable to proprietary Flipkart seller data once a sharing framework is established.
- Generated five actionable feature-level insights to improve seller evaluation checklists immediately.
- Developed a three-tier risk banding framework (Green/Yellow/Red) with a CRM integration architecture and retraining feedback loop.
- Provided industry validation through a Flipkart Enterprise Sales Manager, grounding the methodology in real operational constraints.

7.3 Limitations

- Proxy dataset may not fully capture seller-specific distributional complexity or feature noise.
- Real seller failure is a spectrum (low GMV, poor ratings, partial engagement) rather than a clean binary outcome.
- Critical seller signals - product quality, catalogue depth, geographic demand - are unavailable in the proxy.
- RF probability scores require calibration before deployment to ensure interpretability as true probabilities.
- The IBM dataset lacks temporal structure; seasonal patterns and cohort effects cannot be modelled.

7.4 Future Research

- Apply identical pipeline to Flipkart's proprietary historical seller onboarding data once available.
- Integrate SHAP values for instance-level, human-readable seller risk explanations.
- Implement rolling window validation to simulate real deployment under distributional drift.
- Benchmark XGBoost, LightGBM, and CatBoost against the final RF model for precision improvement.
- Design a controlled A/B pilot study measuring the model's causal impact on seller success rates.
- Develop uplift models to identify which high-risk sellers, with targeted support, would convert to success.



8. References (APA 7th Edition)

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9. Appendix - Proof of Industry Interaction and Data Vetting

9.1 Industry Professional Profile

Field	Details
Name	Saurabh Singh
Designation	Enterprise Sales Manager - Seller Acquisition
Organisation	Flipkart (Walmart Group)
Experience	8+ Years in Enterprise Seller Onboarding
Interaction Objective	Validate problem framing and proxy dataset approach
Interaction Mode	WhatsApp Group Video Call + Follow-up messages
Date	March 2025
Participants	Aman Kumar Singh, Ankit Kumar, Arush Pratap Singh, Sourav Singh

9.2 Industry Expert Photograph

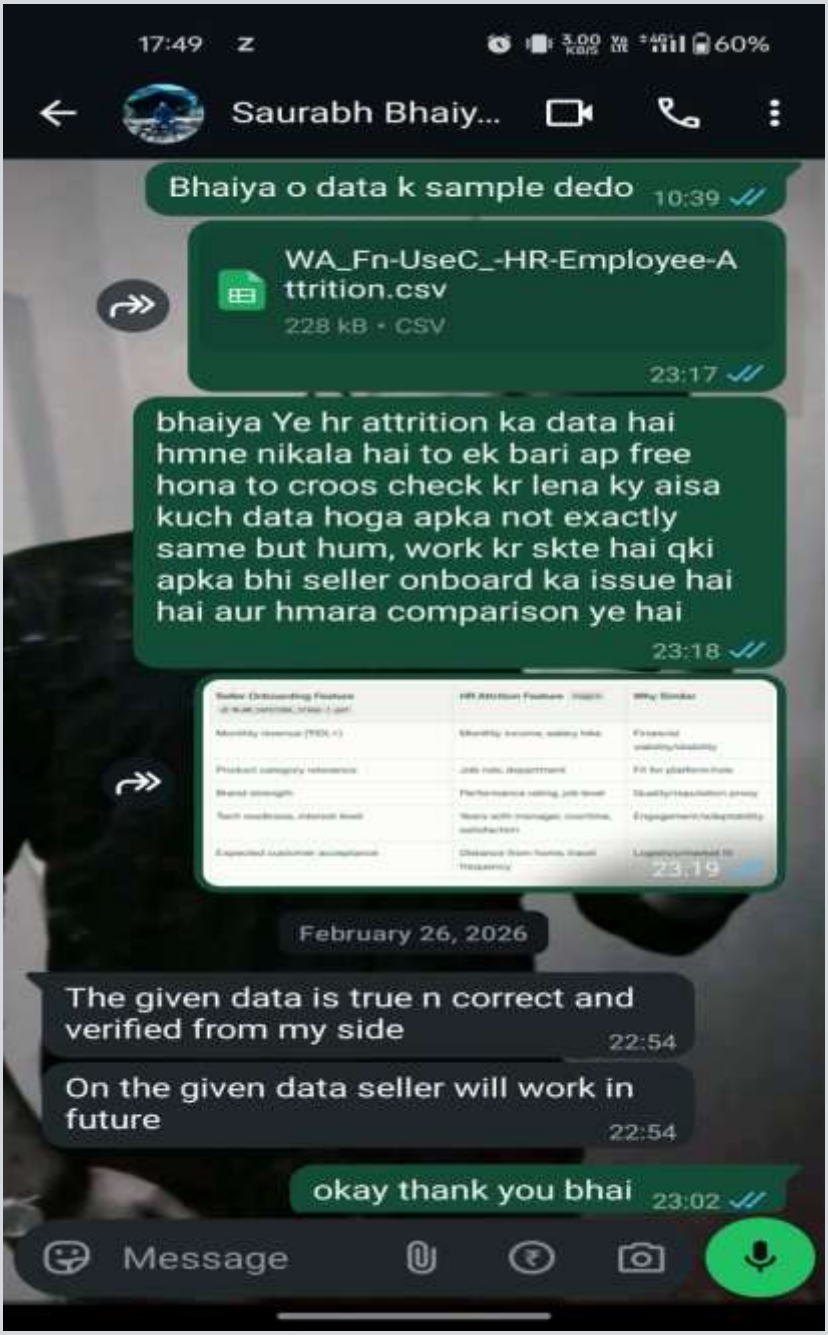
[PHOTO PLACEHOLDER - Photograph of Mr. Saurabh Singh, Enterprise Sales Manager, Flipkart]



9.3 WhatsApp Conversation - Data Request and Validation

The screenshots below show the WhatsApp conversation in which the IBM HR Attrition dataset was shared and the structural equivalence was confirmed by the industry expert:

[PHOTO PLACEHOLDER - WhatsApp Screenshot 1 - Dataset sent by Saurabh Singh with structural comparison table visible]



9.4 Video Call Evidence

[PHOTO PLACEHOLDER - WhatsApp Video Call Screenshot - All 4 team members + Saurabh Singh visible in group call]



9.5 Kaggle Dataset Vetting

The screenshot below confirms the IBM HR dataset was accessed from Kaggle (public domain) with dataset dimensions and URL visible:

[PHOTO PLACEHOLDER - Kaggle Dataset Page - IBM HR Analytics Employee Attrition & Performance (n=1,470, 35 features, public domain)]

